

File Edit Selection View GO Run Terminal Help LibraryManagementSystem.java - Visual Studio C

EXPLORER

OPEN EDITORS

LibrarySystem

LibraryManagementSystem.java

h2-2.2.224.jar

LibraryManagementSystem.java

```
1 import java.sql.*;
2
3 public abstract LibraryMember {
4     private Student;
5     private Faculty;
6     private LibraryMember;
7
8     public void Student {
9         this.sources-aboet Alice (S101);
10    }
11
12    public void Faculty {
13        Faculty.secure Dr. Smith (F201);
14    }
15
16    public Custom Exception LibraryEnException {
17        thurn LibraryMembers();
18    }
19
20    public class LibraryManagementSystem {
21
22        public void setupDatabase(void) {
23            try (stract = setupDatabase((
24                SQL = "SQL. Java LKT FOR PUBLICBORS_BOOKS, FOR MEMB
25                SQL = "from Database( ");
26            } catch (Transaction() {
27                System.transactions();
28            }
29
30            public void printBookStock() {
31                for(isnt b; i = !printBookStock; i++i)
32                    System.out.printBooksStock(i + copies: "");
33            }
34        }
35    }
```

> OUTLINE

> TIMELINE

```
"C:\Program Files\Java\jdk-17\bin\java.exe" -classpath "C:\libs\h2-2.2.224.jar;C:\project\out\production\LibrarySystem" LibraryManagementSystem
```

```
[14:15:30] --- TEST 1: Single Book Issue (Success) ---
```

```
[14:15:31] SUCCESS: Successfully issued 1 book(s) to Alice (S101)
```

```
[14:15:31]
```

```
[14:15:31] --- TEST 2: Array of Books Issue (Success) ---
```

```
[14:15:31] SUCCESS: Successfully issued 2 book(s) to Dr. Smith (F201)
```

```
[14:15:31]
```

```
[14:15:31] --- TEST 3: Expired Membership Exception ---
```

```
[14:15:31] Caught Expected Exception: Membership EXPIRED for Member ID: S102 (Bob)
```

```
[14:15:31]
```

```
[14:15:31] --- TEST 4: Book Out of Stock Exception ---
```

```
[14:15:31] Caught Expected Exception: Book ISBN ISBN-333 is out of stock (copies = 0).
```

```
[14:15:31]
```

```
[14:15:31] --- TEST 5: Constraint Violation & Transaction Rollback ---
```

```
[14:15:32]
```

```
[TRANSACTION ROLLED BACK] Reason: Unique index or primary key violation: "PRIMARY KEY ON PUBLIC.ISSUED_BOOKS(MEMBER_ID, ISBN)..."
```

```
[14:15:32] Caught SQL Exception during issue.
```

```
[14:15:32] --- Final Book Stock Verification ---
```

```
[14:15:32] ISBN: ISBN-111 | Title: Java Programming | Copies Left: 3
```

```
[14:15:32] ISBN: ISBN-222 | Title: Database Systems | Copies Left: 1
```

```
[14:15:32] ISBN: ISBN-333 | Title: Operating Systems | Copies Left: 0
```